

Ahmed Emad - Technical Projects Report

Senior Backend Developer & Automation Engineer *GitHub:*

<https://github.com/ahmedemadm90> | *Email:* ahmedemadm90@gmail.com | *Mobile /*
WhatsApp: +201019030515

Executive Summary

This technical summary outlines the core production builds and architectural systems designed and implemented by Ahmed Emad. The portfolio spans enterprise Laravel multi-tenant SaaS platforms, high-throughput backend ticketing and hospitality systems, Flutter-based IoT and offline mobile applications, and n8n workflow automation pipelines that eliminate repetitive manual operational tasks by 40-60%.

Core Technical Stack & Capabilities

- **Backend Architecture:** Laravel, PHP, MySQL, RESTful APIs, Database Optimization, RBAC.
 - **Mobile & Cross-Platform:** Flutter, Dart, Firebase Push Notifications, Local SQLite Storage.
 - **Automation & Integration:** n8n Workflow Orchestration, Webhooks, Python Scripts, API Integrations.
 - **DevOps & Delivery:** Linux Production Servers, Git Version Control, Docker Containerization.
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Selected Distinct Production Builds

1. Car Rental Marketplace SaaS (`car-rental-marketplace-saas`)

- **Category:** Laravel / Marketplace SaaS
- **Stack:** PHP, Laravel, MySQL
- **Description:** Enterprise multi-tenant car rental marketplace SaaS platform featuring advanced booking lifecycle management, multi-branch fleet tracking, and secure payment processing hooks.

2. Support Ticketing System (tickets)

- **Category:** Operations / Laravel
- **Stack:** PHP, Laravel, JavaScript
- **Description:** Professional support platform built around ticket lifecycle tracking, secure backend routing, role-based agent permissions, and real-time status updates.

3. Hotel Reservation System (hotel)

- **Category:** Hospitality / Workflow
- **Stack:** PHP, Laravel, SCSS
- **Description:** Advanced reservation and property management system designed to organize guest bookings, room availability matrices, and daily hospitality operations.

4. School Management ERP (school-management-system)

- **Category:** Education / ERP
- **Stack:** PHP, Laravel, ERP Architecture
- **Description:** Enterprise resource planning platform built for educational institutions, structured around student admissions, grade records, timetables, and administrative workflows.

5. AutoScan Pro OBD-II Diagnostics (autoscan-pro)

- **Category:** Mobile / IoT / Flutter
- **Stack:** Dart, Flutter, Bluetooth ELM327
- **Description:** Professional OBD-II car diagnostic mobile application featuring ELM327 Bluetooth hardware integration, real-time vehicle telemetry gauges, and an offline simulator mode.

6. SayDone Offline Voice AI Assistant (saydone-offline)

- **Category:** Mobile / AI / Dart
- **Stack:** Dart, Flutter, SQLite
- **Description:** Privacy-focused offline voice-to-task AI assistant equipped with local SQLite database storage, full Arabic/English bilingual support, and configurable daily limits.

7. AutoOps n8n Automation Hub (autoops-n8n-hub)

- **Category:** Automation / n8n
- **Stack:** n8n, Webhooks, Python
- **Description:** Comprehensive n8n workflow repository for automated CRM data synchronization, incident webhook alert routing, and cloud operations pipelines.

8. Ghost Assist Real-Time AI Interviewer (ghost-assist-ai)

- **Category:** Mobile / AI Real-Time
- **Stack:** Dart, Flutter, AI Audio Streaming
- **Description:** Real-time AI-powered assistant featuring live voice listening and stealth mobile UI designed for professional interviews and meetings.

9. Cemex Logistics Archive (logistics)

- **Category:** Operations / Logistics
- **Stack:** PHP, Laravel, Archive Engine
- **Description:** Enterprise logistics archive project structuring dispatch records so operational teams can audit, search, and act on supply chain data with zero latency.

10. Newsmes Content Management System (newsmes)

- **Category:** Content / CMS
- **Stack:** PHP, JavaScript, CMS Architecture
- **Description:** Modern news and publication content management system featuring dynamic categorization, media handling, and structured editorial publishing workflows.

Professional Career Timeline

- **CEMEX Egypt** (Oct 2017 – Present) | *Senior Backend Developer*: Engineered enterprise Laravel apps serving 1,000+ users, processing 10K-30K daily API requests, and optimizing SQL execution by 25-35%.
- **Exception Tech Group** (Jan 2017 – Present) | *Mobile Developer (Flutter)*: Built cross-platform mobile apps connected to Laravel REST backends, serving 500+ active users.

- **Contract / Internal Projects** (2019 – Present) | *Automation & Integration Engineer (n8n)*: Orchestrated automated data pipelines, cutting manual tasks by 40-60%.
 - **Ibnsina Pharma** (Nov 2015 – Dec 2016) | *Backend Developer*: Developed core PHP backend services and internal business tooling APIs.
 - **TE Data** (Oct 2014 – Oct 2015) | *Backend Developer*: Contributed to platform components, debugging, and database reliability.
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Report generated from verified GitHub repository records and professional resume data.